

Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 10



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Recap of Previous Lecture



Topic

Topic

Topic

Regular Languages

Topics to be Covered



Topic

Topic

Topic

Regular Languages

2 min

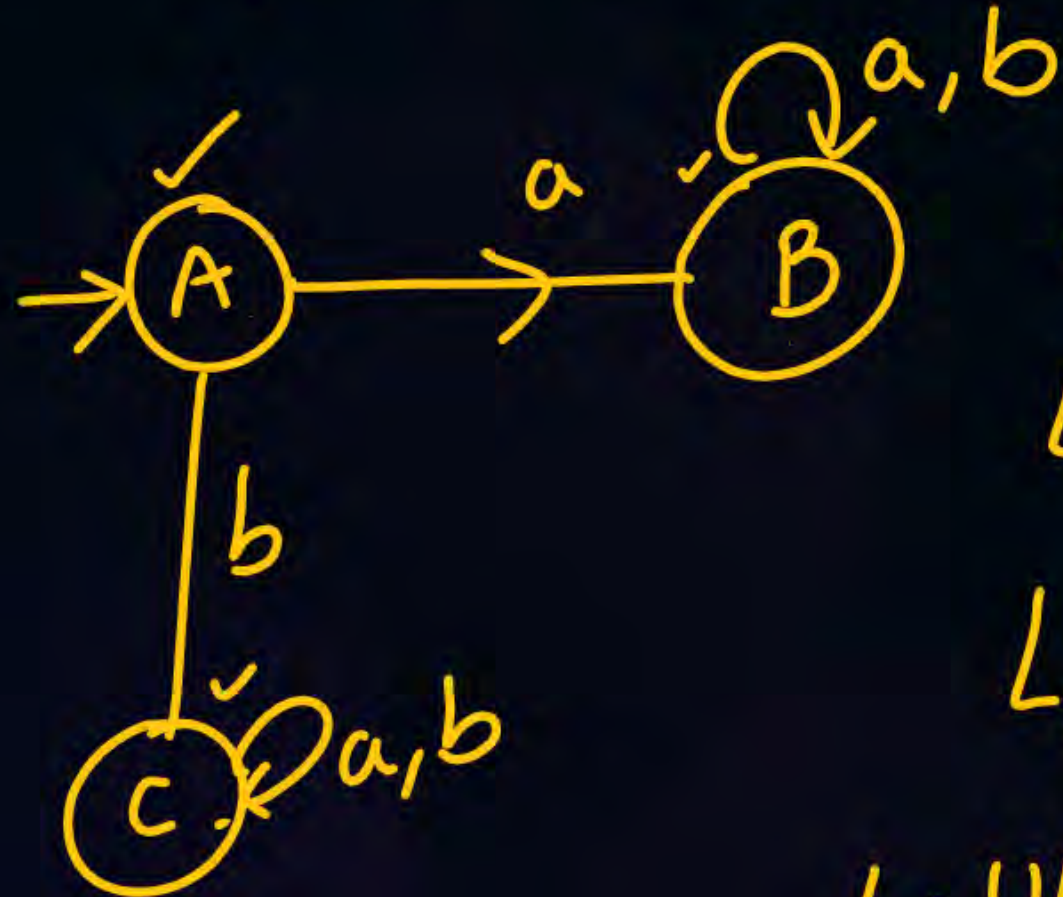
not able to see the comments

myhill-Nerode theorem:-

Based on min DFA

$$\Sigma^* = \{\epsilon, a, b, ab, ba, aa, bb, \dots\}$$

equivalence classes $\rightarrow \underline{L_A, L_B, L_C}$



$$L_A = \{\epsilon\}$$

$$L_B = \{a, aa, ab, aaa, \dots\}$$

$$L_C = \{b, ba, baa, \dots\}$$

$$\underline{L_A \cup L_B \cup L_C = \Sigma^*} \quad L_A \cap L_B \cap L_C = \phi$$

if the equivalence classes are finite, then we can build a FA.

if a language is regular then myhill nirode equivalence classes are finite

Reg $\rightarrow$ min DFA $\rightarrow$ Every state is a equivalence class
no of states are finite

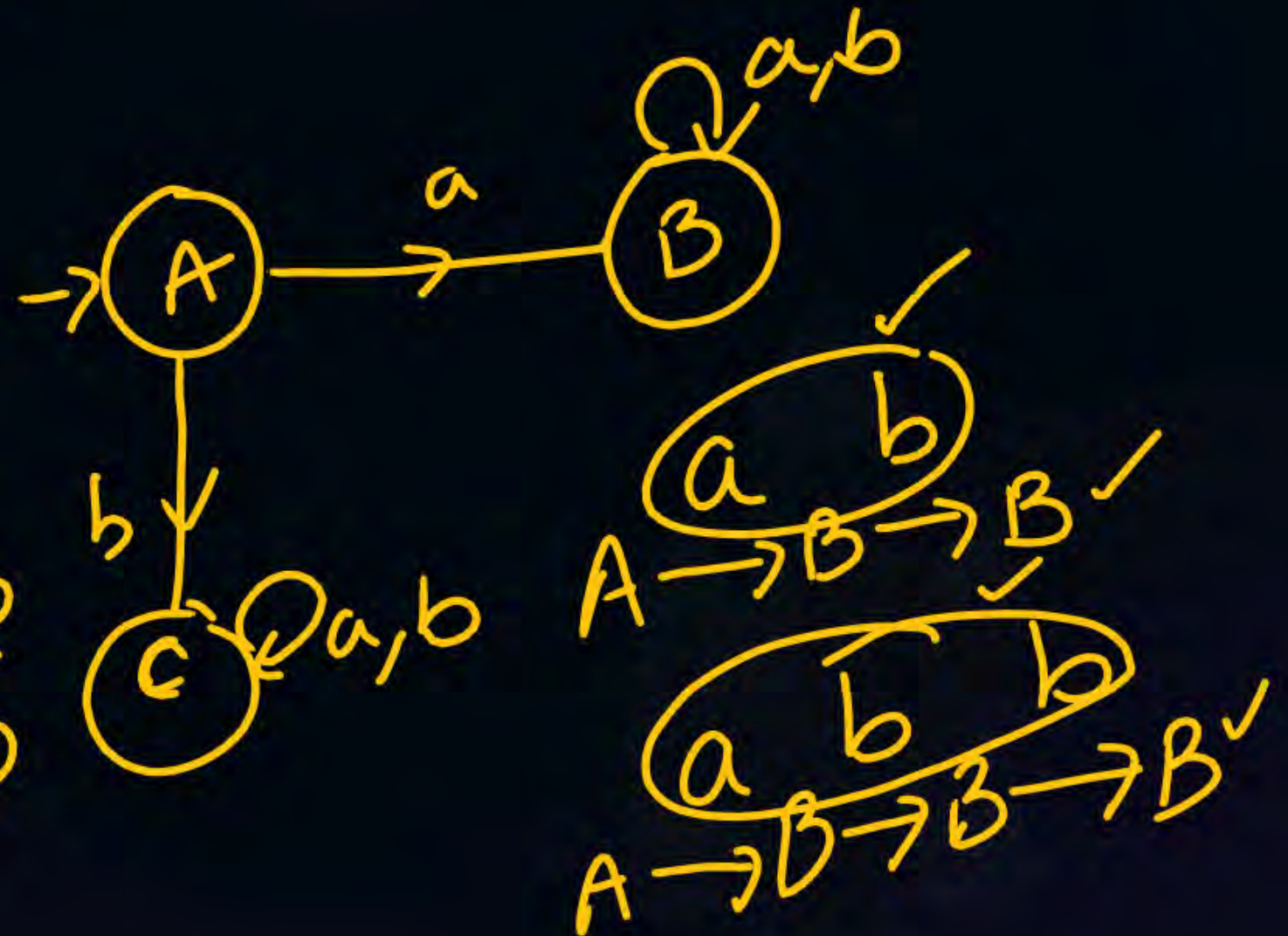
Two string x, y are equivalent.

$$\text{if } \delta^*(q_0, \underline{x}) = \delta^*(q_0, \underline{y})$$

Starting from initial state,

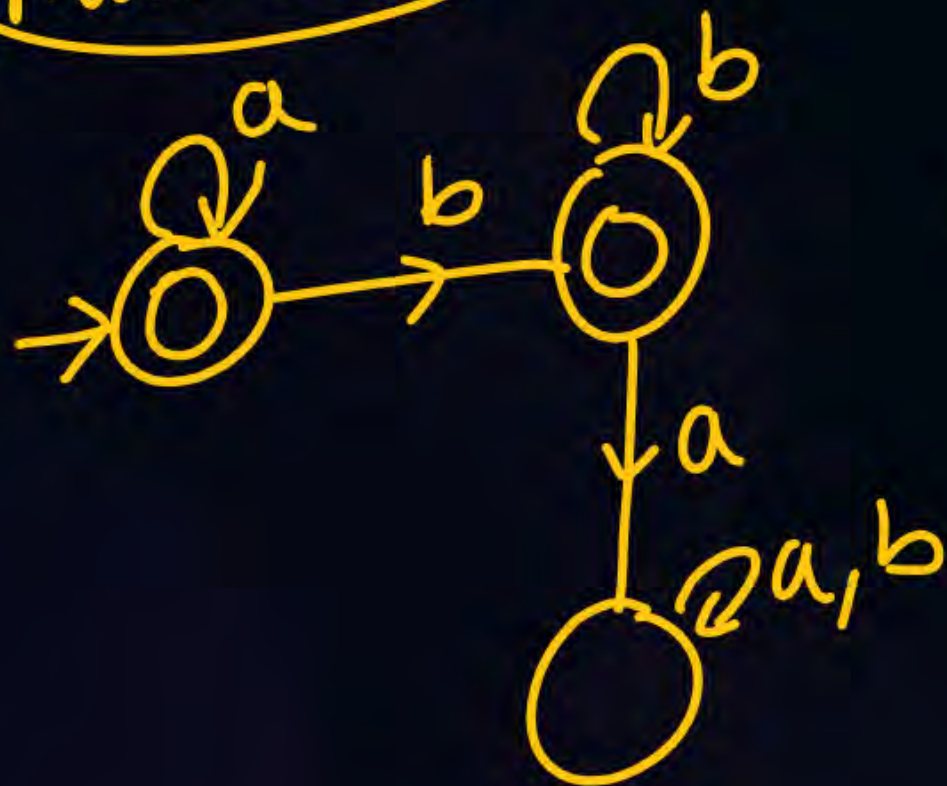
both go to same state

$$\frac{ab \approx abbb}{ba \approx \underline{bbb}}$$



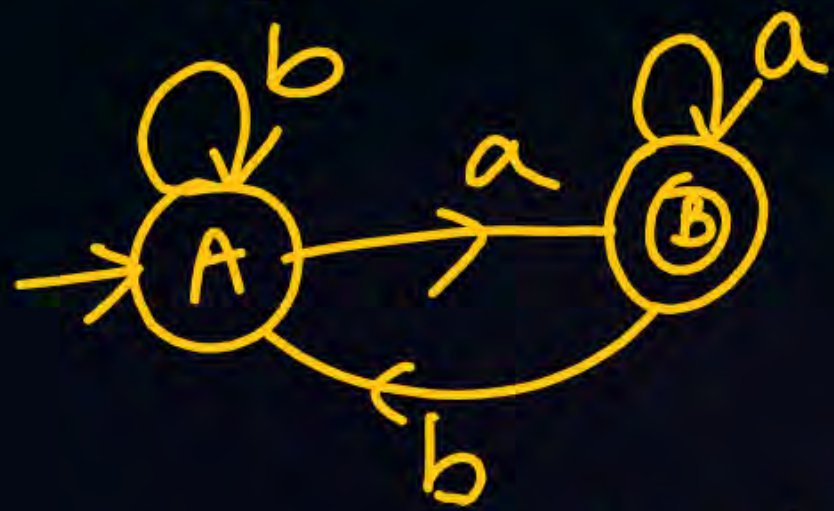
How many mn equivalence classes are in a^*b^*

min DFA



mn equ class - 3

How many MN equi classes are in the language {Set of all strings ending with a 'a'}



= 2

L_B = Set of all strings ending with 'a'

L_A = Set of all strings not ending with 'a'
 $\{ \epsilon \} \cup \{ \text{ending with } b \}$

How many mN equivalence classes are in $a^n b^n / n \geq 1$?

$a^n b^n \rightarrow$ not regular $\rightarrow$ DFA ✓

∞ equivalence classes

→ If $E_1, E_2, \dots, E_n$ are m equi classes

1) $|E_i| \geq 1$ (at least one string will be present)

2) If $i \neq j \Rightarrow E_i \cap E_j = \emptyset$

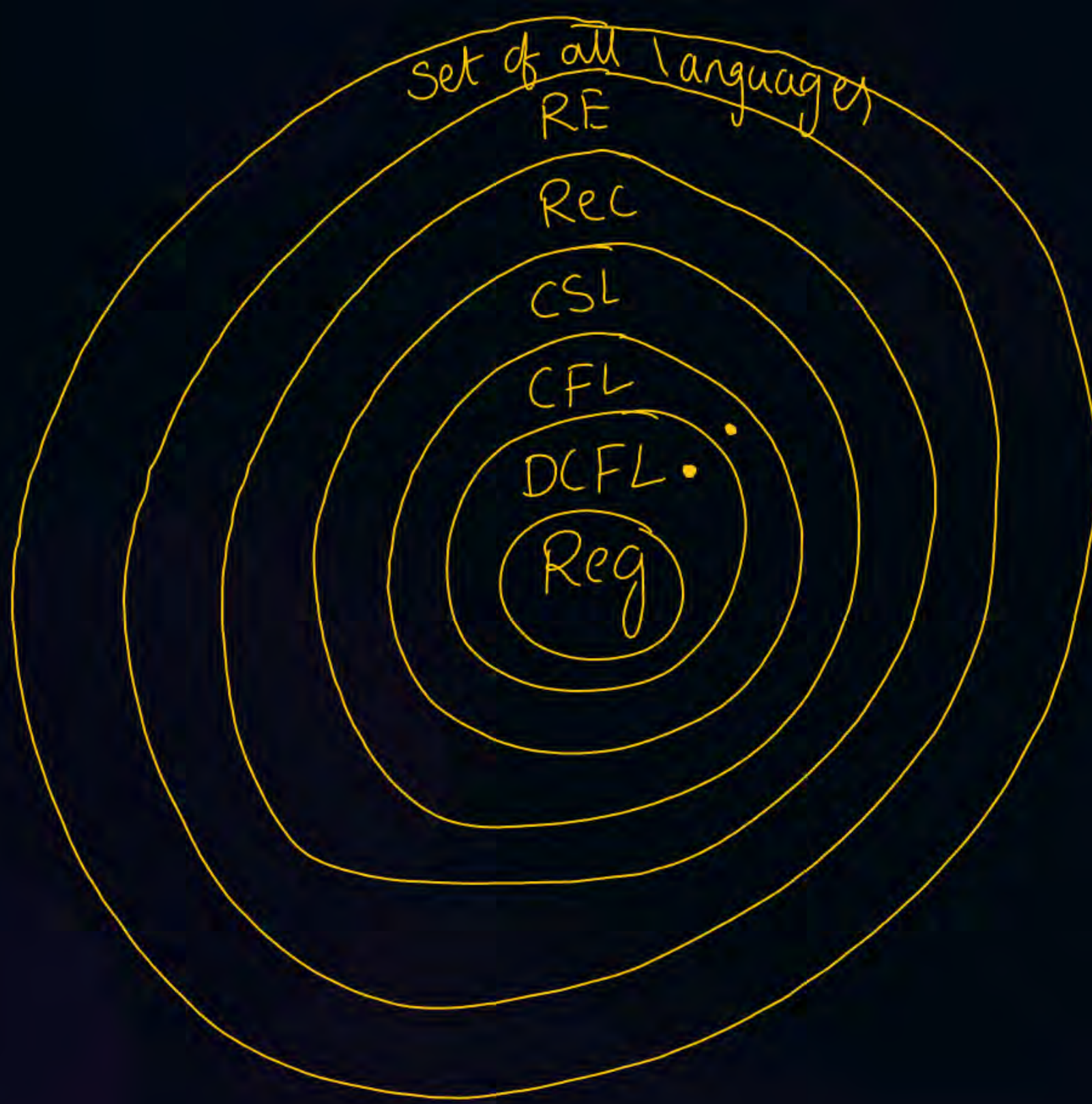
(A string will not be present in two equi classes)

3) $\bigcup E_i = \Sigma^*$

Reg ✓

CFL

CSL



Closure properties



	Reg	DCFL	CFL	CSL	Rec	RE
$\cup$	✓	X	✓	✓	✓	✓
$\cap$	✓	X	X	✓	✓	✓
L^c	✓	✓	X	✓	✓	X
$*$	✓	X	✓	✓	✓	✓
$\cdot$	✓	X	✓	✓	✓	✓
L^R	✓	X	✓	[Crossed out]	[Crossed out]	[Crossed out]
$h(L)$	✓	X	✓			
$h^{-1}(L)$	✓	✓	✓			

$L_1 \rightarrow \text{Reg}$

$L_2 \rightarrow \text{Reg}$

$L_1 \cup L_2 \rightarrow \text{Reg}$

$L_1, L_2 \rightarrow \text{DCFL}$

$\uparrow \text{DCFL} \cup \text{DCFL} \uparrow$

$L \rightarrow \text{may or may not be DCFL}$

$\text{CFL} \cup \text{CFL} \rightarrow \text{CFL}$

$\text{DCFL} \cup \text{DCFL} \rightarrow \text{CFL}$

Closure properties



	Reg	DCFL	CFL	CSL	Rec	RE
$\cup$	✓	x	✓	✓	✓	✓
$\cap$	✓	x	x	✓	✓	✓
L^c	✓	✓	x	✓	✓	x
$*$	✓	x	✓	✓	✓	✓
$\cdot$	✓	x	✓	✓	✓	✓
L^R	✓	x	✓	✓	✓	✓
$h(L)$	✓	x	✓	✓	✓	✓
$h^{-1}(L)$	✓	✓	✓	✓	✓	✓

$\overline{\text{CFL}} \rightarrow$ may or may not be CFL

$\overline{\text{CSL}} \rightarrow \text{CSL}$

$(a^n b^n) c^m \cap a^n (b^m c^m)$

$\uparrow$ DCFL
 $\uparrow$ CFL
 CSL

$\cap$ DCFL $\uparrow$
 $\cap$ CFL $\uparrow$
 $\cap$ CSL $\rightarrow$ CSL

Closure properties



	Reg	DCFL	CFL	CSL	Rec	RE
$\cup$	✓	X	✓	✓	✓	✓
$\cap$	✓	X	X	✓	✓	✓
L^c	✓	✓	X	✓	✓	X
$*$	✓	X	✓	✓	✓	✓
$\cdot$	✓	X	✓	✓	✓	✓
L^R	✓	X	✓			
$h(L)$	✓	X	✓			
$h^{-1}(L)$	✓	✓	✓			

which of the languages
are ^{not} closed under union?

- a) Reg b) DCFL c) Rec
d) RE

Closure properties



	Reg	DCFL	CFL	CSL	Rec	RE
$\cup$	✓	X	✓	✓	✓	✓
$\cap$	✓	X	X	✓	✓	✓
L^c	✓	✓	X	✓	✓	X
$*$	✓	X	✓	✓	✓	✓
$\cdot$	✓	X	✓	✓	✓	✓
L^R	✓	X	✓	[Crossed out]	[Crossed out]	[Crossed out]
$h(L)$	✓	X	✓			
$h^{-1}(L)$	✓	✓	✓			

$(DCFL)^*$

CFL ✓

Closure properties

	Reg	DCFL	CFL	CSL	Rec	RE
$\cup$	✓	X	✓	✓	✓	✓
$\cap$	✓	X	X	✓	✓	✓
L^c	✓	✓	X	✓	✓	X
$*$	✓	X	✓	✓	✓	✓
$\cdot$	✓	X	✓	✓	✓	✓
L^R	✓	X	✓			
$h(L)$	✓	X	✓			
$h^{-1}(L)$	✓	✓	✓			

RE - $\text{Rec} \uparrow$ ✓

RE - RE

RE $\cap$ $\overline{\text{RE}}$ → may or may not be RE

RE - Rec

RE $\cap$ $\overline{\text{Rec}}$

RE $\cap$ $\text{Rec} \uparrow$

RE $\cap$ RE

$\text{RE} \uparrow$ ✓



→ No language is closed under following operations

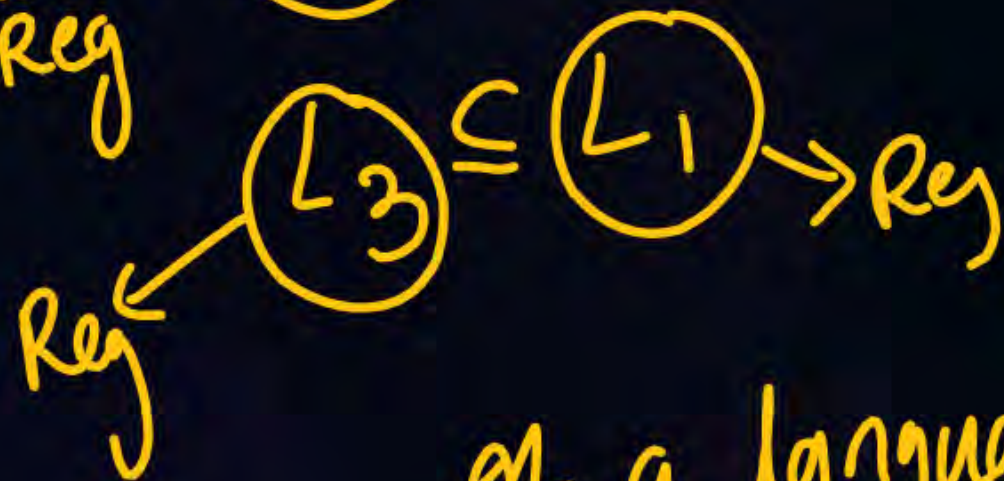
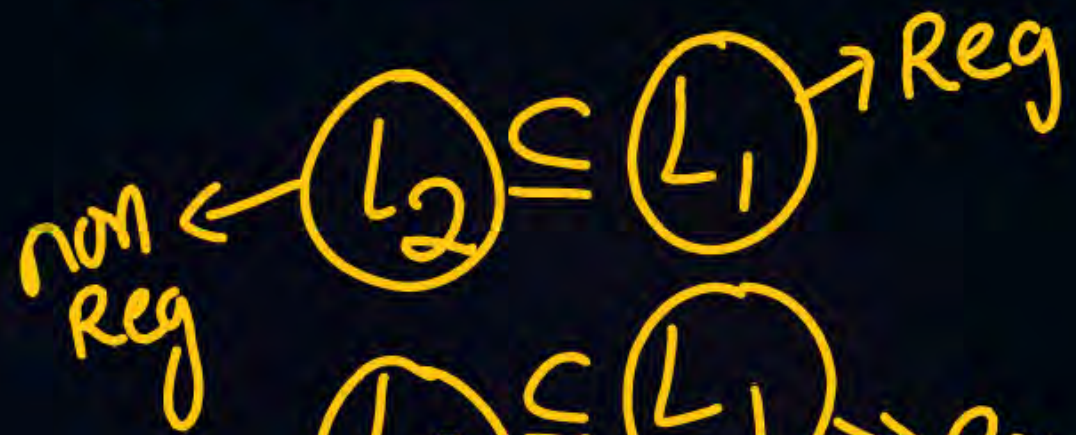
$\subseteq$
 $\supseteq$
 infinite union ✓
 infinite intersection ✓
 infinite set diff → Reg language

Subset of Reg language → may or may not be Reg

Superset of DCFL → may or may not be DCFL

$$L_3 = a^+ b^+$$

$$L_1 = a^* b^* \quad L_2 = a^n b^n$$



if a language is regular then its subset may or may not be reg

$$L_1 = \phi \rightarrow \text{Reg}$$

$$\text{Reg} \leftarrow \phi \subseteq a^n b^n \rightarrow \text{non Reg}$$

$$\text{Reg} \leftarrow \phi \subseteq a^* b^* \rightarrow \text{Reg}$$

if a language is reg, then its superset may or may not be regular

infinite union

$$L = L_1 \cup L_2 \cup L_3 \cup L_4 \cup L_5 \cup L_6 \dots$$

$ab \quad a^2b^2 \quad a^3b^3 \quad a^4b^4 \quad a^5b^5 \quad \dots$

$$a^n b^n / n \geq 1$$

$a \quad a^2 \quad a^3 \quad a^4 \quad a^5 \dots$

$$a^n / n \geq 1$$

infinite union of reg languages may or may not be reg

infinite intersection

$$L_1 \cap L_2 \cap L_3 \cap L_4 \dots$$

$$\bar{L}_1 \cup \bar{L}_2 \cup \bar{L}_3 \cup \bar{L}_4 \dots$$

may not be regular

$L_1 - L_2 - L_3 - L_4 - L_5 \dots$

$L_1 \cap \bar{L}_2 \cap \bar{L}_3 \cap \bar{L}_4 \cap \bar{L}_5 \dots$

may a many not
be done

Secondary operations:-

$$L_1 - L_2 = L_1 \cap \bar{L}_2$$

$$L_1 \oplus L_2 = (L_1 \cap \bar{L}_2) \cup (\bar{L}_1 \cap L_2)$$

$$L_1 \Rightarrow L_2 = \bar{L}_1 \cup L_2$$

$$L_1 \Leftrightarrow L_2 = (L_1 \cap L_2) \cup (\bar{L}_1 \cap \bar{L}_2)$$

$$L_1 \uparrow L_2 = \overline{L_1 \cap L_2}$$

$$L_1 \downarrow L_2 = \overline{L \cup L_2}$$

	Reg	DCFL	CFL	CSL	Rec	RE
LUR	✓	✓	✓	✓	✓	✓
LAR	✓	✓	✓	✓	✓	✓
L-R	✓	✓	✓	✓	✓	✓

$$DCFL \cap R \rightarrow DCFL$$

$$CSL \cup R \rightarrow CSL$$

$$Rec - R \rightarrow Rec$$

$$Reg - DCFL$$

$$= Reg \cap \overline{DCFL}$$

$$= \overline{Reg} \cap \underline{DCFL}$$

$$= \textcircled{DCFL}$$

Tight $\leftarrow$ CFL ✓

CFL $\cap$ Reg $\uparrow$

CFL $\cap$ CFL

CSL ✓

$\downarrow$
Loose

	Reg	DCFL	CFL	CSL	Rec	RE
LUR	✓	✓	✓	✓	✓	✓
LAR	✓	✓	✓	✓	✓	✓
L-R	✓	✓	✓	✓	✓	✓

	Reg	DCFL	CFL	CSL	Rec	RE
LUR	✓	✓	✓	✓	✓	✓
LAR	✓	✓	✓	✓	✓	✓
L-R	✓	✓	✓	✓	✓	✓

Reg - Rec

Reg $\cap$ Rec

Reg $\cap$ Rec

Rec ✓

	Reg	CFL	
prefix Δ init(L)	✓	✓	← Suffix
Right quotient (L/L_2)	✓	✓	← left quotient
Cycle(L)	✓	✓	
min(L)	✓	X	
max(L)	✓	X	
Halt(L)	✓	X	
Alt(L)	✓	X	

X

init(L) Δ prefix(L) :-

$$L = \{\underline{10}, \underline{001}\}$$

$$\text{Prefix}(L) = \{\epsilon, 1, \textcircled{10}, \epsilon, 0, 00, \textcircled{001}\} \checkmark$$

ϵ always belongs to $\text{prefix}(L)$

$$L \subseteq \text{prefix}(L) \checkmark$$

$$L = a^*, \text{ init}(L) = ?$$

$$L = \{\epsilon, \underline{a}, \underline{\underline{aa}}, \underline{\underline{\underline{aaa}}} \dots\}$$

$$\text{Init } L = \{\epsilon, a, aaa, aaaa \dots\}$$

a^*

$$L = a^+ \text{ init}(L)$$

$$L = \{ \underline{a}, \underline{aa}, \underline{aaa}, aaaa \dots \}$$

$$\text{init}(L) = \{ \epsilon, a, aa, aaa \dots \}$$

a^*

$$L = (ab)^* \text{ init}(L)$$

ababab

ϵ , a, ab, ab[✓], abab, abab[✓], ababab

$$\underline{(ab)^*} + \underline{(ab)^*} a = \underline{(ab)^*(\epsilon + a)}$$

$$L = \underline{a}^* \underline{b}^* \text{ init}(L)$$

$\epsilon, a, aa, aaa, \dots$

$\epsilon, b, bb, bbb, \dots$

$ab, aab, abb, aabb, \dots$

$$\cancel{a^* + b^*} + a^* b^*$$

$$\underline{\underline{a^* b^*}}$$

$$L = (a+b)^* \quad \text{init}(L)$$

$$\begin{array}{ccc}
 L & \leq & \text{init}(L) \\
 \downarrow & & \downarrow \\
 (a+b)^* & & (a+b)^* \checkmark
 \end{array}$$

$L = a^n b^n / n \geq 1, \text{init}(L)?$

aaabbb

$\epsilon, a, aa, aaa, \underline{aaaa}b, \underline{aaaabb}, \underline{\underline{aaaabbb}}$

$\underline{a^n b^n / n \geq 1} \cup \underline{a^m b^1 / m \geq 1} \cup \underline{\{\epsilon\}}$



$\cancel{a^*} \cancel{b^*} \checkmark$

$$L = \{ w / n_a(w) = n_b(w) \}$$

$\{ \epsilon, \underline{a}b, \underline{b}a, \underline{aabb}, abab, abbaab, \dots \}$

$\epsilon, a, b, ab, ba, aab, \dots \}$

$(a+b)^*$
 $\underline{aba}b$

$\underline{abb}a \in L$

$\underline{\underline{aabb}}bb \in L$

Right quotient:-

$$L_1 = \{ \underline{1101}, \underline{01} \}$$

$$L_2 = \{ \underline{01}, 1, 0 \}$$

$$\begin{array}{ccccc} \cancel{\frac{11}{\cancel{01}}} & \cancel{\frac{110}{1}} & \overset{\phi}{\left(\frac{1101}{0} \right)} & \cancel{\frac{\cancel{01}}{\cancel{01}}} & \cancel{\frac{01}{1}} & \overset{\phi}{\left(\frac{01}{0} \right)} \end{array}$$

$$L_1/L_2 = \{ 11, 110, \epsilon, 01 \}$$

left quotient :-

$$L_1 = \{ \underline{1101}, 01 \}$$

$$L_2 = \{ 01, 1, 0 \}$$

$$\left(\frac{1101}{01} \right) \quad \frac{1101}{x} \quad \left(\frac{1101}{0} \right)$$

$$\underline{\underline{\{ 101, \epsilon, 1 \}}}$$

$$\begin{array}{c} \epsilon \\ \hline \cancel{01} \end{array} \quad \frac{01}{1} \quad \frac{\cancel{01}}{\cancel{0}}$$

$$L_1 = \underline{\underline{10^*}} \quad L_2 = 0 \quad L_1/L_2 = ?$$

$$L_1 = \{1, 10, 100, 1000, 10000, \dots\}$$

$$\times \begin{array}{r} \textcircled{1} \\ \textcircled{0} \end{array} \quad \begin{array}{r} \cancel{10} \\ \cancel{0} \end{array} \quad \begin{array}{r} \cancel{100} \\ \cancel{0} \end{array} \quad \begin{array}{r} \cancel{1000} \\ \cancel{0} \end{array}$$

$$\textcircled{10^*} \checkmark$$

$$a^*/\underline{a}$$

$$\epsilon, a, aa, aaa \dots$$

$$a/a = \epsilon \quad a^2/a = a \quad a^3/a = \underline{a}$$

$$\epsilon, a, aa, aaa \dots = a^*$$

$$a^+/a$$

$$\in \frac{a}{a}, \frac{aa}{a}, \frac{aaa}{a}, \dots$$

$$\epsilon, a, aa, aaa, \dots, a^* \checkmark$$

$$aa^+/a$$

$$\frac{a\phi}{\phi}, \frac{aa\phi}{\phi}, \frac{aaa\phi}{\phi} \dots$$

$$a \quad aa \quad aaa \dots$$

$$\textcircled{a^+}$$

$$L_1 = \frac{10^*}{\cdot} \quad L_2 = 01^* \quad L_1/L_2$$

$$L_2 = \{ \textcircled{0}, \textcircled{01}, \textcircled{011}, \dots \}$$

$$\frac{10^*}{0} = \textcircled{10^*} \quad \textcircled{\frac{10^*}{01}} \quad \textcircled{\frac{10^*}{011}}$$

$\textcircled{10^*}$

$$\frac{10^+}{01^+}$$

$$\frac{10^+}{01}$$

$$\frac{10^+}{011}$$

$$\frac{10^+}{0111}$$

$$\frac{10^+}{01111} \dots$$

$$\textcircled{\phi} \begin{aligned} \text{inf}/\text{inf} &= \text{inf} \text{ of finite} \\ \text{fin}/\text{fin} &= \text{finite} \end{aligned}$$

CFLURL $\uparrow$ CFLUCFL $\rightarrow$ CFL ✓



$$L_1 = \underline{a^n b^n / n \geq 1} \quad L_2 = \underline{(a+b)^*}$$

$$L_1 \cup L_2 = ? \quad (a+b)^* \rightarrow \text{Reg} \checkmark$$

a) Reg b) CFL c) CSL d) Rec



THANK - YOU